

Assignment B

Generating Random Inputs

- ▶ Where applicable, you may use the function shown below to generate input values randomly:

```
float random_data(float low, float hi)
{
    float r = (float)rand() / (float)RAND_MAX;
    return low + r * (hi - low);
}
```

- ▶ For example, you may use this function to populate the input matrices in Assignment 1, and initialize the input values in option pricing assignments.

Measuring Running Time using Chrono

- ▶ We use a timer to measure the execution times of a program.
- ▶ We have many choices to get such timing measurements.
Code snippet below shows how to use chrono in the C++ Standard Library.

```
#include <chrono>

using namespace std::chrono;

int main()
{
    high_resolution_clock::time_point t1 =
        high_resolution_clock::now();

    do_the_works();

    high_resolution_clock::time_point t2 =
        high_resolution_clock::now();

    std::cout << "Elapsed time: " <<
        duration_cast<milliseconds>(t2 - t1).count() << " ms";

}
```

- ▶ Defined in chrono header.

Assignment B (C/C++)

- ▶ Due: March 31 by 2 PM CT.
- ▶ Write a function to multiply two $N \times N$ matrices.
- ▶ Use the function you wrote to multiply two 100×100 matrices.
- ▶ Measure the execution time.
- ▶ There are many ways to represent a matrix. Two options are shown below (next slide).
- ▶ You are not required to use techniques such as vectorization/multithreading for this assignment.
- ▶ The Aim of this assignment is to get the students to think about performance and set the stage for week 2 lecture.
- ▶ As long as anyone makes a genuine and honest attempt to solve this problem one will get full points for this assignment, even if the solution is not complete.

1. Use a third party library

- ▶ Example: Eigen [http:](http://eigen.tuxfamily.org/index.php?title=Main_Page)

[//eigen.tuxfamily.org/index.php?title=Main_Page](http://eigen.tuxfamily.org/index.php?title=Main_Page)

- ▶ Getting started with Eigen is easy: [https:](https://eigen.tuxfamily.org/dox/GettingStarted.html)

[//eigen.tuxfamily.org/dox/GettingStarted.html](https://eigen.tuxfamily.org/dox/GettingStarted.html)

2. Use `std::vector`:

```
using matrix = std::vector<std::vector<float>>>;  
or  
typedef std::vector<std::vector<float>>> matrix;
```

- ▶ Code snippet below shows
 - ▶ how to use a vector to represent a matrix (as defined above)
 - ▶ how to initialize it using random input values

```
int main()
{
    const int rows = 1000, columns = 1000;
    matrix m1;

    //allocate space for matrix elements
    m1.resize(rows);

    for (int i = 0; i < rows; ++i)
    {
        m1[i].resize(columns);
    }

    //populate the matrix using random values
    for (int i = 0; i < rows; ++i)
    {
        for (int j = 0; j < columns; ++j)
        {
            m1[i][j] = random_data(0, 10);
        }
    }
}
```